MECH 309: Assignment 4, Question 3

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Question 1(a)

In the uncontrolled case, $u_k = 0$ for all $k \geq 0$, so the state recursion becomes

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k, \qquad k = 0, 1, 2, \ldots$$

We prove that

$$\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0 \qquad \text{for all } k = 0, 1, 2, \ldots$$

using base cases $k = 0$ and $k = 1$, then induction starting from $k = 1$.

Base case ($k = 0$). Since $\mathbf{A}^0 = \mathbf{I}$,

$$\mathbf{x}_0 = \mathbf{I}\mathbf{x}_0 = \mathbf{A}^0 \mathbf{x}_0.$$

Base case ($k = 1$). Using the recursion with $k = 0$,

$$\mathbf{x}_1 = \mathbf{A}\mathbf{x}_0 = \mathbf{A}^1 \mathbf{x}_0.$$

Inductive step (from $k = 1$). Assume for some $k \geq 1$ that $\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0$. Then

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k = \mathbf{A}(\mathbf{A}^k \mathbf{x}_0) = \mathbf{A}^{k+1} \mathbf{x}_0.$$

Therefore, by induction, $\mathbf{x}_k = \mathbf{A}^k \mathbf{x}_0$ holds for all $k = 0, 1, 2, \ldots$.

Question 1(b)

Assume $\mathbf{A}$ is diagonalizable, so there exists an invertible matrix $\mathbf{V}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{V}\mathbf{D}\mathbf{V}^{-1}.$$

We show that for all integers $k \geq 1$,

$$(\mathbf{V}\mathbf{D}\mathbf{V}^{-1})^k = \mathbf{V}\mathbf{D}^k\mathbf{V}^{-1},$$

and then conclude an explicit form for $\mathbf{x}_k$.

Base case ($k = 1$). Trivially,

$$(\mathbf{V}\mathbf{D}\mathbf{V}^{-1})^1 = \mathbf{V}\mathbf{D}\mathbf{V}^{-1} = \mathbf{V}\mathbf{D}^1\mathbf{V}^{-1}.$$

Inductive step. Assume for some $k \geq 1$ that

$$(\mathbf{V}\mathbf{D}\mathbf{V}^{-1})^k = \mathbf{V}\mathbf{D}^k\mathbf{V}^{-1}.$$

Then

$$\begin{aligned}(\mathbf{VDV}^{-1})^{k+1} &= (\mathbf{VDV}^{-1})^k(\mathbf{VDV}^{-1}) \\ &= (\mathbf{VD}^k\mathbf{V}^{-1})(\mathbf{VDV}^{-1}) \\ &= \mathbf{VD}^k(\mathbf{V}^{-1}\mathbf{V})\mathbf{DV}^{-1} \\ &= \mathbf{VD}^{k+1}\mathbf{V}^{-1},\end{aligned}$$

since $\mathbf{V}^{-1}\mathbf{V} = \mathbf{I}$. Therefore, by induction,

$$(\mathbf{VDV}^{-1})^k = \mathbf{VD}^k\mathbf{V}^{-1} \qquad \text{for all } k = 1, 2, \ldots$$

Equivalently,

$$\mathbf{A}^k = \mathbf{VD}^k\mathbf{V}^{-1} \qquad \text{for all } k = 1, 2, \ldots$$

Finally, combining this with part (a), where $\mathbf{x}_k = \mathbf{A}^k\mathbf{x}_0$, we obtain

$$\boxed{\mathbf{x}_k = \mathbf{VD}^k\mathbf{V}^{-1}\mathbf{x}_0, \qquad k = 0, 1, 2, \ldots}$$

(where the case $k = 0$ also holds since $\mathbf{D}^0 = \mathbf{I}$ and thus $\mathbf{VD}^0\mathbf{V}^{-1} = \mathbf{I}$).

Question 1(c)

Assume the eigenvalues of $\mathbf{A}$ are real and distinct. Then $\mathbf{A}$ is diagonalizable and we may write

$$\mathbf{A} = \mathbf{VDV}^{-1}, \qquad \mathbf{D} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

From part (b),

$$\mathbf{x}_k = \mathbf{VD}^k\mathbf{V}^{-1}\mathbf{x}_0.$$

Since $\mathbf{V}$ and $\mathbf{V}^{-1}$ do not depend on k, the limit $\mathbf{x}_k \to \mathbf{0}$ as $k \to \infty$ is governed by $\mathbf{D}^k$. But

$$\mathbf{D}^k = \operatorname{diag}(\lambda_1^k, \lambda_2^k, \ldots, \lambda_n^k).$$

Thus $\mathbf{D}^k \to \mathbf{0}$ as $k \to \infty$ if and only if $\lambda_i^k \to 0$ for each i, which holds exactly when

$$|\lambda_i| < 1 \qquad \text{for all } i = 1, 2, \ldots, n.$$

Therefore, a sufficient (and in this diagonalizable setting, equivalent) condition for $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{0}$ is

$$\boxed{|\lambda_i| < 1 \ \text{ for all } i = 1, 2, \ldots, n}.$$

Question 1(d)

Consider the controller

$$u_k = -\mathbf{K}\mathbf{x}_k, \qquad \mathbf{K} \in \mathbb{R}^{1\times n}.$$

Substituting into

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}u_k$$

gives

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}(-\mathbf{K}\mathbf{x}_k) = (\mathbf{A} - \mathbf{BK})\mathbf{x}_k.$$

Hence the closed-loop (controlled) state matrix is

$$\boxed{\bar{\mathbf{A}} = \mathbf{A} - \mathbf{BK}}.$$

Question 1(e)

Under the feedback in part (d), the closed-loop dynamics are

$$\mathbf{x}_{k+1} = \bar{\mathbf{A}}\mathbf{x}_k, \qquad \bar{\mathbf{A}} = \mathbf{A} - \mathbf{B}\mathbf{K}.$$

Using the conclusion of part (c), a sufficient design requirement on $\mathbf{K}$ is to choose it so that every eigenvalue μ_i of $\bar{\mathbf{A}}$ satisfies

$$|\mu_i| < 1 \qquad \text{for all } i = 1, 2, \ldots, n.$$

Equivalently,

$$\boxed{\text{choose } \mathbf{K} \text{ such that } \operatorname{eig}(\mathbf{A} - \mathbf{B}\mathbf{K}) = \{\mu_i\}_{i=1}^{n} \text{ and } |\mu_i| < 1 \ \forall i}.$$

Question 1(f)

The following numerical experiments were performed using the provided Python code. In each case, we computed eigenvalues using `numpy.linalg.eigvals` and plotted the state components q_k and v_k versus time.

(i) No control ($u_k = 0$)

With $u_k = 0$, the update is $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$. The eigenvalues of $\mathbf{A}$ computed numerically are

$$\operatorname{eig}(\mathbf{A}) = [1,\ 0.99501248] \quad \text{(from Python output)}.$$

We check the condition from part (c):

$$\text{Condition from (c):} \quad |\lambda_i| < 1 \ \ \forall i.$$

From the numerical eigenvalues, this condition is *not satisfied* (since $|\lambda_1| = 1$), so the observed open-loop response is *inconsistent* with $\mathbf{x}_k \to \mathbf{0}$.

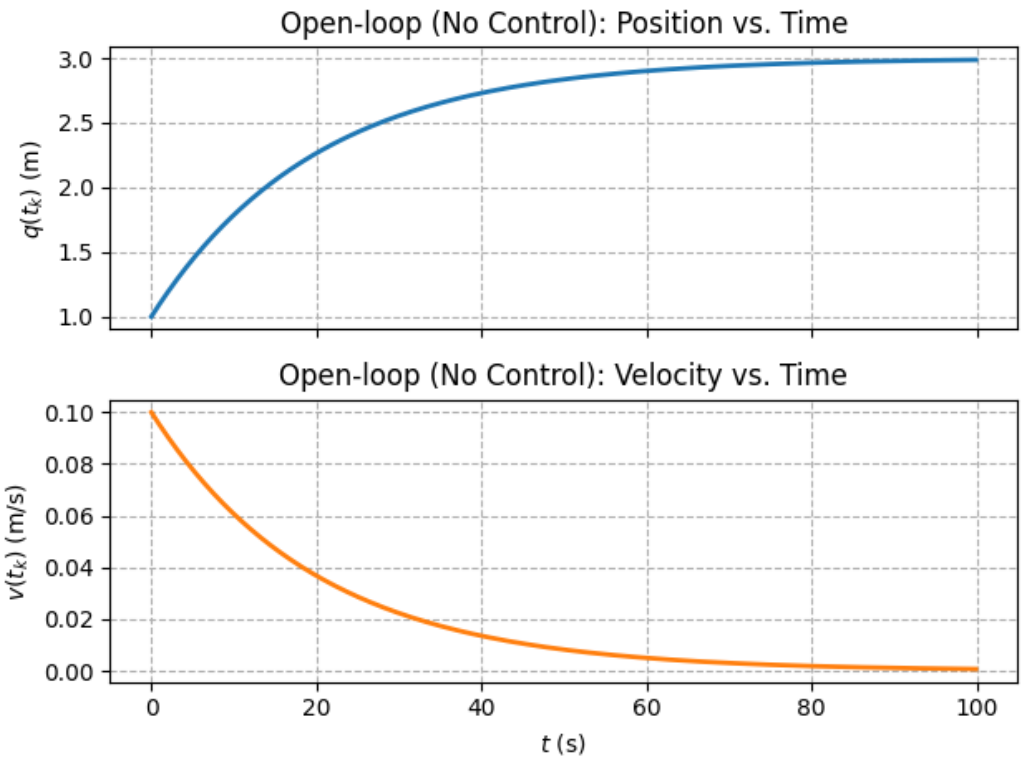

Figure 1: Open-loop response ($u_k = 0$): q_k and v_k versus time.

(ii) With control ($u_k = -\mathbf{K}\mathbf{x}_k$)

With the controller $u_k = -\mathbf{K}\mathbf{x}_k$, the update becomes

$$\mathbf{x}_{k+1} = \bar{\mathbf{A}}\mathbf{x}_k, \qquad \bar{\mathbf{A}} = \mathbf{A} - \mathbf{B}\mathbf{K}.$$

The eigenvalues of $\bar{\mathbf{A}}$ computed numerically are

$$\text{eig}(\bar{\mathbf{A}}) = [0.90999965,\ 0.89000055] \quad \text{(from Python output)}.$$

We again apply the criterion from part (c):

$$\text{Stability requirement:} \quad |\mu_i| < 1 \ \ \forall i.$$

For the provided $\mathbf{K}$, this condition is *satisfied*. The plot shows both components decaying toward zero, matching the prediction that $\mathbf{x}_k \to \mathbf{0}$.

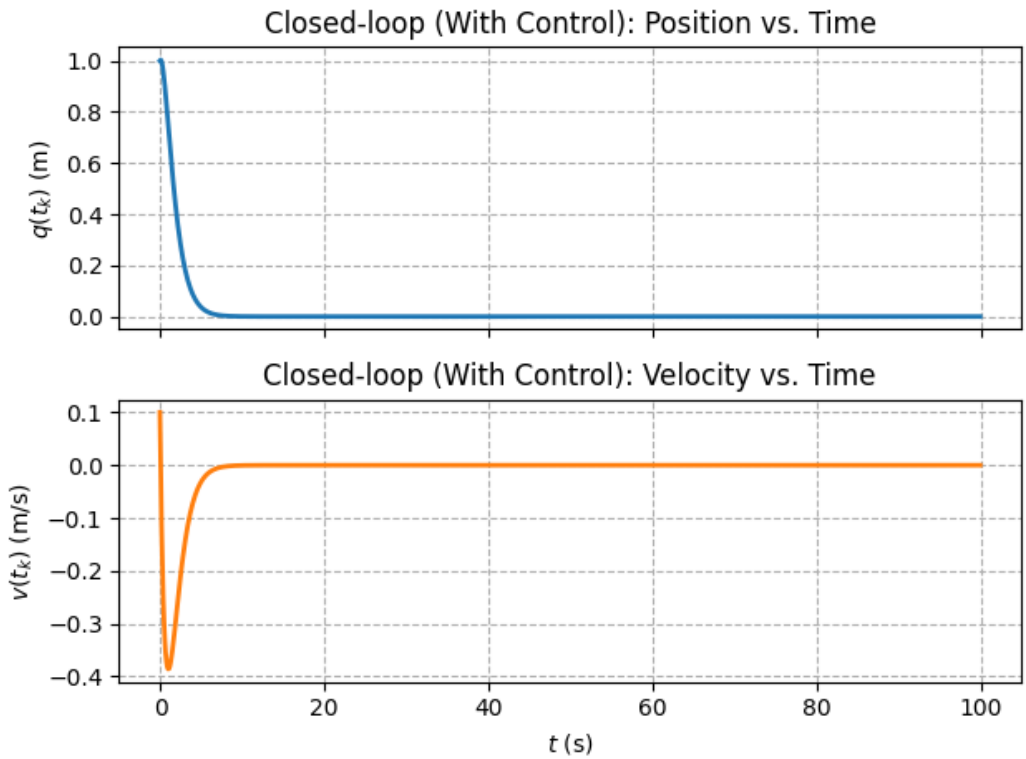

Figure 2: Closed-loop response ($u_k = -\mathbf{K}\mathbf{x}_k$): q_k and v_k versus time.